

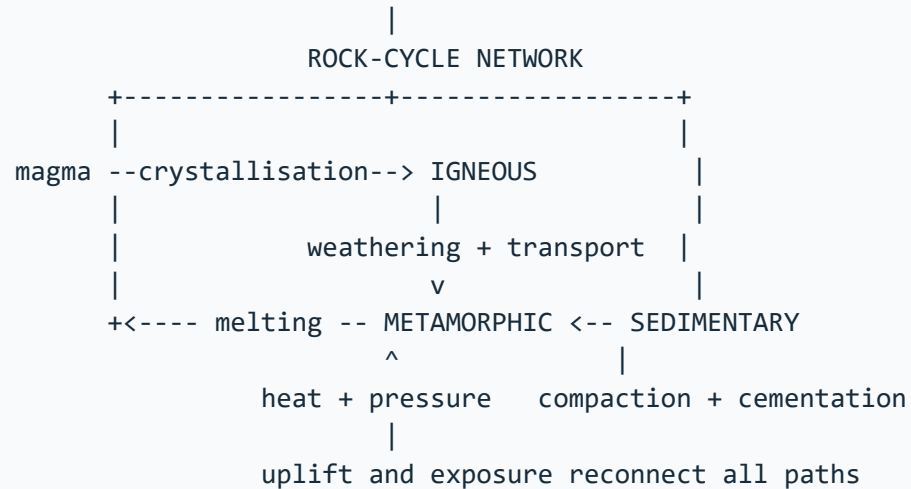
ASCII MASTER FLOW – PANEL 1/12: Crust, lithosphere and the rock-cycle network

```
+----- EARTH MATERIAL SYSTEM -----+
| mineral -> rock -> lithology -> petrology -> geology |
+-----+
```

Crust = compositional shell

Lithosphere = crust + rigid uppermost mantle

Asthenosphere = weaker mechanical layer below



The cycle is a network: no rock family has one compulsory next stage.

MUST REMEMBER: Crustal analysis connects seismic evidence, compositional and rheological layers, rock cycle, plate boundaries and Wilson-cycle logic to India's cratons, mobile belts, Gondwana basins, Deccan volcanics, Himalaya and Indo-Gangetic foredeep.

ASCII MASTER FLOW – PANEL 2/12: Igneous, sedimentary and metamorphic diagnostic matrix

+-----+-----+-----+		
FAMILY	PROCESS	FIELD CLUES
+-----+-----+-----+		
Igneous	magma/lava cooling	crystals, glass, vesicles
intrusive	slow cooling	coarse texture
extrusive	fast cooling	fine or glassy texture
Sedimentary	deposit + lithify	beds, fossils, ripples
Metamorphic	solid-state change	grade, new minerals
foliated	differential stress	aligned fabric
+-----+-----+-----+		

Composition pairs:

granite-rhyolite = felsic | diorite-andesite = intermediate

gabbro-basalt = mafic

FIREWALLS

bedding is not foliation | loess and moraine are deposits until lithified

porosity is void space | permeability is connected flow capacity

ASCII MASTER FLOW – PANEL 3/12: From continental drift to plate tectonics and Wilson-cycle closure

WEGENER'S DRIFT CLAIM

```
|
+ -> shelf-edge fit      + -> matching rocks and fossils
+ -> palaeoclimate       + -> later palaeomagnetic support
```

limitation: no adequate driving mechanism

SEA-FLOOR SPREADING

ridge creation -> magnetic stripes -> older/deeper floor outward

```
|
+----- trench destruction ---+
```

PLATE TECTONICS

+	+	+
divergence	subduction	transform
creates crust	destroys oceanic	conserves crust
ridge/rift	trench + arc	shallow quakes

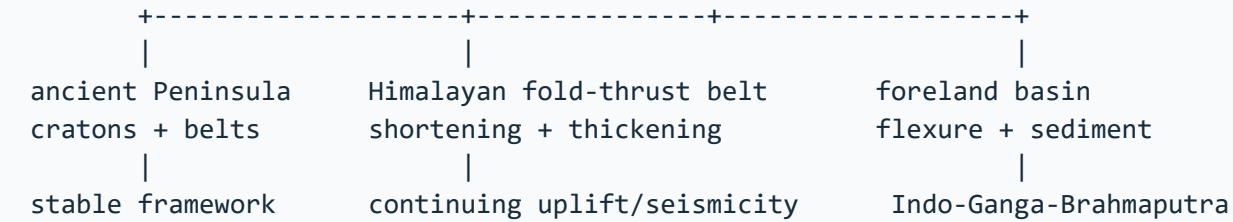
rift -> narrow sea -> ocean -> subduction -> collision -> suture

Wilson cycle is an analogy, not a compulsory destiny for every ocean.

ASCII MASTER FLOW – PANEL 4/12: Indian Plate journey and the three-part geological skeleton

Gondwana rifting -> northward Indian Plate drift -> Tethys closure

|
India-Eurasia collision



NORTH-SOUTH EXAM CROSS-SECTION

Tethyan / Greater / Lesser / Siwalik Himalaya -> foreland alluvium
-> Peninsular cratons, rifts, basins and Deccan cover

EVIDENCE LADDER

superposition -> cross-cutting -> unconformity -> fossil correlation
-> radiometric event age -> palaeomagnetism -> geophysics

Stable Peninsula means relatively stable, not uniform, flat or aseismic.

ASCII MASTER FLOW – PANEL 5/12: Indian rock systems, basins and resource associations

SYSTEM	INDIAN EXPRESSION	SAFE ASSOCIATION
Archaean/Dharwar	gneiss + supracrustals	process-specific ore
Proterozoic	Cuddapah, Vindhyan	stone; bounded ores
Gondwana	faulted inland basins	coal, not universal
Marine basins	coasts and offshore	petroleum potential
Deccan Trap	stacked flood basalts	soil, stone, water

GONDWANA BASINS: Damodar | Son-Mahanadi | Satpura | Godavari

PETROLEUM SYSTEM:

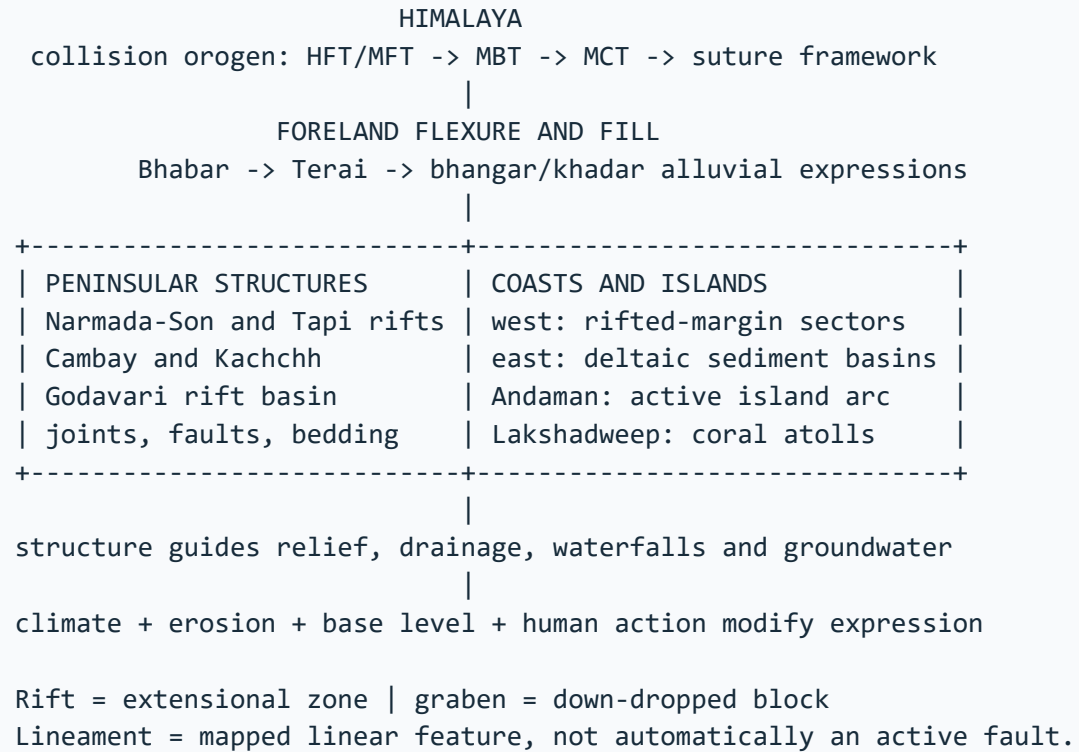
source -> maturation -> migration -> reservoir -> seal -> trap -> timing

DECCAN ARCHITECTURE:

flow top / vesicles / fractures / weathered contact / dense flow interior

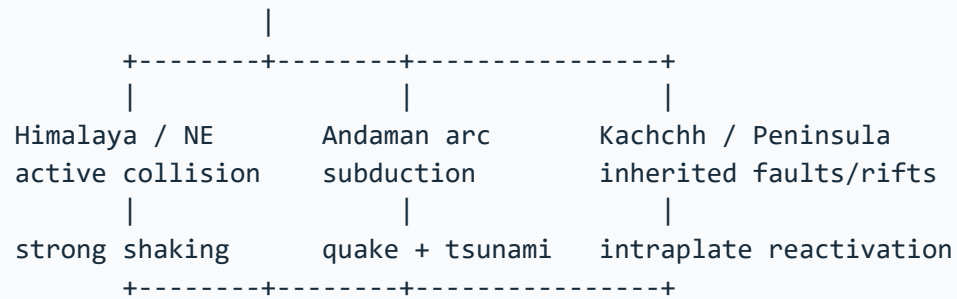
Association is not determinism: grade, preservation, exploration,
technology, economics and environmental limits decide usable resources.

ASCII MASTER FLOW – PANEL 6/12: Himalaya, foreland, coasts, islands and structural controls



ASCII MASTER FLOW – PANEL 7/12: India seismic expression and geology-to-risk translation

TECTONIC AND INHERITED STRUCTURE



SITE FILTER: alluvium, basin depth, groundwater, slope, construction

EXPOSURE + VULNERABILITY - CAPACITY

DISASTER RISK

BIS Zones II-V are broad design generalisations, not predictions.
Geological association must be followed by site evidence and engineering.

RESPONSE: map -> investigate -> code -> inspect -> retrofit -> prepare.

ASCII MASTER FLOW – PANEL 8/12: Map-answer and Mains spine for India's geological architecture

QUESTION: How does geological structure shape India's surface geography?

1. DEFINE: age + composition + rock architecture + tectonic structures

2. DRAW: Himalaya -> foreland plain -> Peninsular shield cross-section

3. SEQUENCE: Gondwana rift -> drift -> collision -> uplift -> sediment fill

4. MAP NAMED UNITS

cratons | Purana basins | Gondwana basins | Deccan | coasts | islands

5. LINK PROCESS TO CONSEQUENCE

structure -> relief/drainage/water/resource/seismic expression

6. QUALIFY

association is not reserve; stable is not aseismic; alluvium has structure

7. VERDICT

inherited geology is filtered through tectonics, climate, erosion and use

Answer order: thesis -> map -> evidence -> mechanism -> India example

-> limitation -> graded conclusion.

ASCII MASTER FLOW – PANEL 9/12: Evidence ladder and confidence control

EVIDENCE LADDER

1. = from source book = inference / standard knowledge = current affairs.
2. Sedimentary characteristics: derived from pre-existing rock (igneous...
3. Quartzite Sandstone Hard; source rock easily identified

VERDICT: Source type controls the strength and limits of the historical claim.

ASCII MASTER FLOW – PANEL 10/12: Examiner traps and contested boundaries

CLOSE DISTINCTIONS

1. Evidence tags: [FACT] directly retrieved or officially verified...
2. Causal method Initial condition -> energy/force or driver ->...
3. [LIMIT] No volatile reserve percentage, unsupported plate rate,...

VERDICT: Qualification earns marks; false certainty is a hard failure.

CLOSE DISTINCTION: Crust is compositional while lithosphere is mechanical; continental crust is generally granitic and less dense, oceanic crust basaltic and denser, and a shield, craton, platform, basin and plate are not interchangeable map units.

ASCII MASTER FLOW – PANEL 11/12: Integrated answer spine and qualified conclusion

ANSWER SPINE

1. Learner-v2 generation g2 Part A - Physical Geography Prelims GS-I +...
2. Practice contract Every solved item has demand decoding, a detailed...
3. Geological Survey of India/PIB - Field Season 2025-26 conclusion and...

VERDICT: Claim, named evidence, analysis and qualification lead to a graded...

EVIDENCE LIMIT: Present relief is a multi-stage tectonic-denudational product; rock age, orogeny, mineralisation and surface form need not coincide, so geological association supports probability rather than deterministic resource claims.

EVIDENCE LADDER

1. = from source book = inference / standard knowledge = current affairs.
 2. Sedimentary characteristics: derived from pre-existing rock (igneous...
 3. Quartzite Sandstone Hard; source rock easily identified
- VERDICT: Source type controls the strength and limits of the historical claim.